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Airfare prediction: Leveraging market data for better decision-making

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ABSTRACT

Airline revenue management is crucial for airlines to maintain their competitive position in the market. Revenue management addresses two main concerns in airline planning processes, pricing and seat inventory management, to balance supply and demand. Pricing or determination of airfare is a complex decision-making process influenced by factors including distance, number of passengers, market share, competition, and route-related characteristics. However, it is a central element as it impacts revenue generation, market positioning, demand management, cost recovery, and customer relationships. This study investigates the machine learning perspective on predicting airline market-level airfares and examines the determinants of airfare. In this regard, exploiting the publicly available data from the US Department of Transportation Bureau of Transportation Statistics, several supervised machine learning algorithms are tested and compared to obtain the most effective prediction for the given dataset. The Random Forest model outperformed the other models, with R^2_{adj} and RMSE scores of 0.998 and 1.811, respectively. An ad hoc feature importance analysis is also performed to gain further insight into the determinants of market-level airfares. The findings emphasize the importance of operational costs and pricing strategies in airfare prices.

1. Introduction

As one of the most complex systems in the world, air transportation plays a vital and expanding role in the global economy. Both demand and competition in air travel are increasing. In this highly competitive industry, pricing and revenue management have become essential aspects for airlines to maintain their competitive position in the market. Pricing and revenue management addresses two main concerns in airline planning processes, that is determining the fare levels (i.e. pricing) and the number of seats to make available at each fare level (i.e. revenue management), to balance supply and demand in such a way that maximizes the revenue. The first step in this process is to define an overall pricing strategy based on costs and competition and to establish the fares and fare rules. Next, on the inventory management side, the demand for each fare class is forecasted, and seat limits for each class are determined (Barnhart, Belobaba, & Odoni, 2003).

Pricing, also referred to as the determination of airfare, is a highly complex decision-making process. That is mainly because of the wide variety of factors that impact airfare, such as distance, number of passengers, market share, competition, route-related characteristics, and overall economic conditions (Belobaba, Odoni, & Barnhart, 2015; Botimer & Belobaba, 1999; Mumbower, Garrow, & Higgins, 2014). Hence, the problem of pricing has been receiving growing attention in the literature (Mantin & Koo, 2009; Stavins, 2001). Over time, the evolution of pricing strategies has been shaped by shifts in regulatory environments, competitive pressures, and changing consumer expectations. During the regulated era of the airline industry airlines operated under strict government regulation, which regulates their fares, routes, and market entry. During that time, fare structures were standardized with little variation between airlines and the pricing was primarily based on distance-based tariffs rather than market-driven factors. The major element of competition was service quality rather than price. This limited competition kept fares relatively high, with little variation. After the Airline Deregulation Act of 1978 in the United States, the air transportation industry experienced significant changes. This act removed fare and route restrictions, leading to increased competition, price wars, and innovations in pricing strategies. As a result, airline pricing became less tied to distance travel, and airlines set fares based on competition, market share, and costs rather than just distance. In post post-regulation era, airlines try to segment demand in each origin-destination (O—D) market by offering various fare products with different price levels and restrictions (Belobaba et al., 2015; Botimer & Belobaba, 1999; Vinod, 2021). From a theoretical perspective, airlines can use three principles when determining fares for each O-D: cost-based, demand-based and service-based pricing. In practice, on the other hand, a mix of these principles is being used. With the rise of low-

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cost carriers (LCCs) offering ultra-low fares, traditional carriers have been forced to compete with discounted options. As competition increases, profit margins tighten, and technology advances, airlines are moving toward more innovative and effective solutions in pricing (Belobaba et al., 2015; Touraine, 2021).

Alongside advancements in pricing strategies, revenue management (RM) has undergone a parallel transformation, from manual booking decisions in the 1970s to sophisticated, data-driven systems. The first computerized RM systems, developed in the early 1980s, initially focused on collecting and storing data from Computer Reservation Systems. Deregulation and technology accelerated this evolution, prompting airlines to adopt advanced database management and analytical methods to manage the increasing complexity of seat inventory control. By the mid-1980s, RM systems included basic monitoring capabilities that allowed actual flight bookings to be tracked in real time relative to expected booking thresholds. By the late 1980s, third-generation RM systems emerged, capable of forecasting and optimizing by booking class for each upcoming flight departure while also retaining the database and booking monitoring features of earlier systems. These systems used historical and real-time booking data to predict demand for each booking class on future flights. They also included overbooking models based on no-show data to calculate optimal overbooking levels (Barnhart et al., 2003). The models and approaches used within these systems also advanced with scientific and technological developments. The earliest RM model was presented by Littlewood in 1972 as a simple decision rule for a two-fare model. This model was then extended to multiple nested fare classes by Belobaba (1987) and Belobaba (1989). The Expected Marginal Seat Revenue (EMSR) model was further enhanced in 1992, evolving into EMSRb, which employs heuristic decision rules for seat allocation across fare classes (Belobaba, 1992). Initially, focused on single-leg flights, RM shifted to network RM (i.e. O—D control) systems, which improve on leg control by managing availability at the O—D level, considering both the market and itinerary (instead of just legs). Virtual Nesting, first introduced by American Airlines in the 1980s as an initial O—D control strategy, involves assigning each itinerary or fare type to a hidden "virtual" class within the airline's reservation system. In the 1990s, to achieve O—D control, some large airlines relied on leg-based EMSR values to approximate the revenue contribution of connecting itineraries (Belobaba, 1998). Then, Displacement-Adjusted Virtual Nesting (DAVN) is introduced, enhancing virtual nesting by adjusting availability to account for the network revenue displacement of connecting passengers, allowing more flexible and optimized seat allocation (Barnhart et al., 2003; Belobaba et al., 2015; Poelt, 2016).

As of the 2020s, airlines still largely depend on traditional methods—using preset fares and ancillaries along with booking class inventory control—to develop and manage their offers. However, there is a gradual shift away from these older practices by adopting the The International Air Transport Association (IATA) New Distribution Capability (NDC). NDC enables airlines to quote any fare from a continuous range rather than being restricted to fixed classes. These new developments, along with the availability of technology, brought dynamic offers, which rely on dynamic and continuous pricing, to the forefront of airline pricing and RM. While dynamic pricing has been around since the 1980s, the major advancement today is the ability to adjust prices in real-time without third-party systems and without being limited to the traditional booking classes, allowing airlines to better align prices with demand. Continuous pricing builds on dynamic pricing by enabling airlines to offer an unlimited range of price points, allowing for more precise adjustments based on real-time supply and demand (2021; Szymański, Belobaba, & Papen, 2021; Long & Belobaba, 2024).

Though dynamic offer generation is still in its infancy, the discussions and research adopting it effectively are gaining momentum. In applying continuous pricing, understanding of competitors' fares and customers' willingness to pay has become crucial as competition significantly influences fare structures and price levels in each market. Airlines often match competitors' fare offerings, particularly the lowest-price competitors, including specific price levels and restrictions. Not matching a competitor's fare or restrictions often leads to a loss of market share, but in some cases, it may have a less negative revenue impact. At this point understanding the type of market, the market concentration, the type of competitors and its effect on fares gained more importance. The use of data and pricing models allows airlines to make better, context-driven decisions, leading to optimized revenue generation (Belobaba et al., 2015; Datalex, 2024). Today, with the existing data and computational capabilities, traditional pricing and RM methods can leverage emerging machine-learning techniques to enhance decision-making (Kumar, Li, & Wang, 2018). There is a growing body of research in leveraging data and machine learning in airline pricing.

Among the studies on airline pricing, recently, airfare prediction has been of particular interest. Studies on airfare prediction can be categorized into two groups based on the target variable and features considered: specific itinerary-level and market segment-level prediction (i.e. predicting airfare for a single flight leg within an O—D route). An airline market is defined as an O—D city pair, which is achieved either by a single flight leg or a route that consists of multiple legs. Specific itinerary-level prediction focuses on forecasting the price of a particular flight, on a specific day, for a specific departure time and airline—essentially the exact flight a customer might book. These models typically incorporate dynamic features such as the number of days until departure, day of the week, and other flight-specific details (Abdella, Zaki, Shuaib, & Khan, 2021). Market segment-level prediction, on the other hand, deals with predicting the average airfare within a broader O—D city pair by mainly taking into account various O—D characteristics, such as historical average prices, competition, route distance, and demand patterns (Wang et al., 2019). However, as O—D control (i.e. Network Revenue Management) is becoming increasingly important for airlines, it is critical for them to be able to predict airfare trends at the market level. This capability enables airlines to adjust their strategies and allocate resources effectively (Barnhart et al., 2003).

Studies on airfare prediction, whether at the specific itinerary-level or market segment-level, employ a variety of techniques ranging from conventional statistical techniques, such as linear regression, to different kinds of advanced data mining and machine learning techniques. Among these approaches, machine learning (ML) is significantly more prominent in recent studies, as it facilitates the inference of rules and model variations on this sophisticated problem. These aspects are important in identifying the most effective and efficient solution and enhancing the prediction performance.

A significant number of these studies use either private datasets or datasets that are constructed through web crawling on online booking sites. Wang et al. (2019), underline the difficulty in extending the work done with private datasets, as it is problematic to gain access to these datasets (Wang et al., 2019). They also mention that crawled data represents only a small portion of ticket sales and has a tendency to be inaccurate in representing the true nature of an entire market. Consequently, they recommended a framework where publicly available databases provided by the United States Bureau of Transportation Statistics (BTS) are used. They consider, as they call it, a market segment-level prediction problem, which refers to a single flight leg within an O—D route. In this study, we also benefit from the publicly available datasets provided by BTS and leverage ML techniques. While the aforementioned study proposed a market-segment level airfare prediction framework, our study differs from it in the sense of the target variable and the features. This study considers a market-level (i.e. O—D level) prediction problem, regardless of whether the route is direct or includes multiple legs. This reflects a more comprehensive, network-aware approach that aligns with airline revenue management goals. Accordingly, a different set of features are considered within the scope of this study to better reflect market-related determinants of air fare. Our O—D level prediction framework achieved a more effective and efficient solution by adopting a different approach to defining the target variable as well as incorporating a broader range of features for prediction.

The remainder of this paper is organized as follows. Section 2 presents a brief overview of the studies on airfare prediction. Section 3 describes the data preparation process and the ML algorithms used in the scope of this study. Section 4 provides the results obtained, along with a discussion. Finally, Section 5 concludes with a summary and future research directions.

2. Literature review

As mentioned in the previous section, airfare prediction is a highly complex problem, mainly due to the variety of factors that impact airfare. However, it is important to have accurate models for determining airfare as it can provide a competitive advantage and maximize revenue. As complex as airline pricing strategies are for airlines, they make sense from an economic perspective, yet they can still be confusing for customers. Accordingly, for over a decade, there has been a growing interest in this problem in the literature. Studies on airfare prediction can be categorized into two groups in terms of the scope of the prediction: prediction at the level of specific itinerary and market segment. Furthermore, there are two main approaches within the literature, based on purpose: those focused on maximizing airline revenue (airline-oriented) and those aimed at assisting consumers in deciding the ideal time to purchase tickets (customer-oriented). A brief summary of recent studies on airfare prediction can be seen in Table 1.

Specific itinerary-level prediction focuses on predicting airfares for complete trips, aims to provide insights into general airfare trends for entire journeys. Within these studies, Domínguez-Menchero et al. (2014) and Groves and Gini (2015) focus on ideal purchase time from a customer-oriented approach. Domínguez-Menchero et al. (2014) introduced a method to examine how airlines price their tickets, aiding consumers in deciding the optimal time for purchase. They employed non-parametric isotonic regression methods to identify the timeframe within which consumers can delay purchasing without facing significant price hikes, assess the economic impact of delaying each day, and determine instances where waiting until the last day is advantageous (Domínguez-Menchero et al., 2014). The analysis focused on four routes between city pairs over a two-month period, encompassing both direct and one-stop flights. Daily price fluctuations were scrutinized for 30 days leading up to the travel date, with prices derived from the lowest fare identified through a specific website's search engine. Groves and Gini (2015) introduced an algorithm for optimizing the timing of airline ticket purchases by predicting the future anticipated lowest prices across all available flight options using ML methods, such as PLS regression, decision tree, and nu-SVRRidge Regression. The algorithm uses a feature-selection approach to capture temporal relationships within the data and reduce the number of features. They used the data they collected from a travel search website. The data covers 7 different O—D pairs. They split the dataset into 3 with the following lengths: 48, 20, and 41 days. They used these sets as the training set, calibration set, and test set, respectively. The authors mentioned that the algorithm approached the optimal purchasing policy far more closely than other decision-theoretic methods in the same field (Groves & Gini, 2015). Vu et al. (2018) proposed a model to help air passengers predict airfare trends by uncovering trends as well as actual airfare changes up to the departure dates without official information from airlines. The authors built a model by stacking two separate prediction models (Random Forest and Multilayer Perceptron) to predict ticket prices, using public airfare data available online. The authors developed a scraping tool to collect airfare data from websites selling tickets to study an emerging aviation market. They chose the Vietnamese aviation market as a case study due to its rapid growth. The model outperformed the Multilayer Perceptron and Random Forest (Vu et al., 2018).

There are also studies at the specific itinerary-level, that focus on the price prediction based on an airline-oriented approach, investigating the factors affecting the fare prices. Lantseva et al. (2015) analyzed the Russian air transportation market and compared the behavior of prices on local and global flights departing from Moscow and Saint-Petersburg during a specific period of time. Two different ticket price information aggregators were used as data sources. The authors developed an empirical, data-driven regression model for predicting airfare across various flight routes (Lantseva et al., 2015). Tziridis et al. (2017) studied a specific itinerary-level airfare prediction problem using machine learning techniques. They collected data from the web manually. The data consisted of 1814 one-way flights on a single O—D pair (from Thessaloniki to Stuttgart). The features that affect the airfares were investigated. The authors selected eight state-of-the-art regression ML

Table 1
Summary of studies on airfare prediction.

Reference	Problem	Dataset	Techniques used	Performance results
Domínguez-Menchero, Rivera, and Torres-Manzanera (2014)	Specific Itinerary-level, Custumer-Oriented	Collected 2 months of data for 4 O-Ds	Non-parametric isotonic regression	N/A
Groves and Gini (2015)	Specific itinerary-level, Custumer-Oriented	Collected 109 days of data for 7 O-Ds	PLS regression, decision tree and nu-SVRRidge Regression	69 % of the optimal savings with PLS regression
Lantseva, Mukhina, Nikishova, Ivanov, and Knyazkov (2015)	Specific itinerary-level, Airline-Oriented	Obtained 4 months of data for 4 origins from two different ticket price information aggregators	Regression using sigmoid and cubic parabola functions	N/A
Tziridis, Kalampokas, Papakostas, and Diamantaras (2017)	Specific itinerary-level, Airline-Oriented	Collected data from the web manually for a single O-D	Multilayer Perceptron, Generalized Regression Neural Network, Extreme Learning Machine, Random Forest Regression Tree, Regression Tree, Bagging Regression Tree, Regression SVM, Linear Regression	88 % accuracy (% - MSE between the desired and predicted prices)
Vu, Minh, and Phung (2018)	Specific itinerary-level, Custumer-Oriented	Collected 3 months of data for 5 single leg O-Ds	Stacked prediction model based on Random Forest and Multilayer Perceptron	4.4 % and 7.7 % better performance compared to Random Forest and Multilayer Perceptron respectively based on R^2 score
Wang et al. (2019)	Market segment-level Prediction, Airline-Oriented	Obtained data from DB1B and T-100 databases, included macroeconomic data	Linear Regression, SVM, Multilayer Perceptron, XGBoost, Random Forest	Adjusted R^2 score of 0.869 with Random Forest
Zhao, You, Gan, Li, and Ding (2022)	Specific itinerary-level, Airline-Oriented	Obtained 2 years for more than 200 O—Ds from an airline (+1.7 million data pieces)	Multi-Attribute Dual-stage Attention, CNN-LSTM, CNN-LSTM+Attention, Auto-Regressive Integrated Moving Average, XGBoost, Random Forest	Multi-Attribute Dual-stage Attention performed at least 2.3 % better thane other models (RMSE: 0.01489±0.003, MAE: 0.01134±0.004)

models and achieved 88 % accuracy (i.e. % - MSE between the desired and predicted prices) for a certain type of flight features (Tziridis et al., 2017). Zhao et al. (2022) proposed an airfare prediction approach featuring a Multi-Attribute Dual-stage Attention mechanism, which integrates diverse data types extracted from a unified dimension. The dataset was a two-year anonymous airfare record from an airline. The authors reported that the proposed approach outperformed the prediction models relying on Auto-Regressive Integrated Moving Average, random forest, or deep learning models with at least 2.3 % better performance (Zhao et al., 2022).

On the market segment-level, Wang et al. (2019), studied an airfare prediction problem combining two publicly available datasets of BTS (Airline Origin and Destination Survey (DB1B) and Air Carrier Statistics (T-100)) with macroeconomic data. From the DB1B database, they used ticket and coupon data (Detailed information on these databases and the data they contain can be found in section 3.1). They calculated the market segment-level fare based on the itinerary fare information in the ticket table and the distance ratio (i.e. the proportion between the distance of each leg in the coupon table and the full length of the itinerary in the ticket table). The authors presented an ML framework, testing five different algorithms. Random Forest was identified to be the best regression model on the testing dataset, with a high performance of 0.869 adjusted R squared score (Wang et al., 2019).

Airfare data is hard to obtain and requires gathering and processing before use, which is a time-consuming process. Most studies in the literature tested their approach on various datasets by crawling the web or requesting private data from companies. As a result, replicating the research and comparing the performance of the models is challenging. In other words, one of the primary challenges for researchers in this field is a lack of benchmarking data. The DB1B and T-100 databases maintained by BTS contain publicly accessible data for US airlines. Researchers, decision-makers, and industry experts utilize the data in the DB1B database to better understand the air transportation industry and identify opportunities for improvement. These databases provide valuable information that can be used in airfare prediction. To this extent, this study utilizes these two databases in presenting a high-performance model for O—D level airfare prediction. To the best of our knowledge, this is the first attempt to present an ML-based framework for O—D (i.e. market) level airfare prediction problems using publicly available data.

The databases, data preparation process, and ML algorithms used in the context of this study are described in the following section.

3. Methodology

In the previous chapter, we defined our problem as market-level airfare prediction. In this section, we present our research methodology by providing detailed information on the databases used, the dataset preparation process, the features selection process, the ML methods we tested, and our experimental setup. The flow chart of our study is presented in Fig. 1.

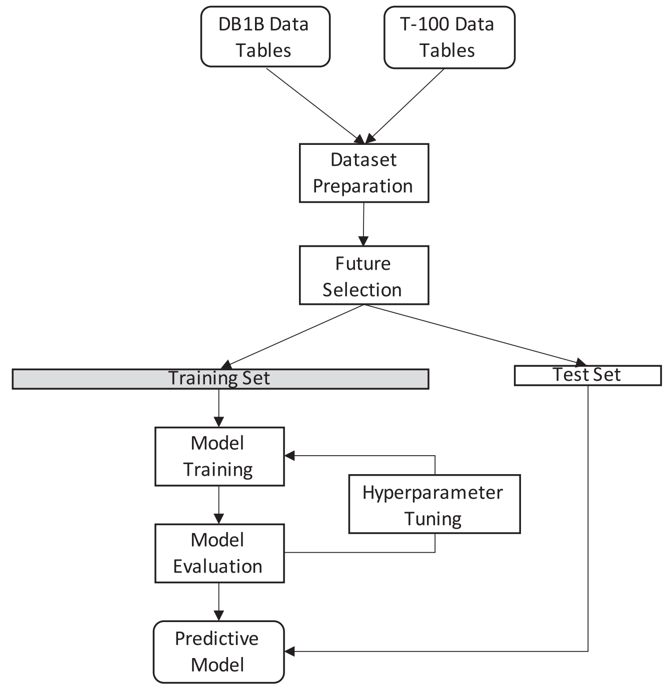

Fig. 1. Study flowchart.

3.1. Databases

As we mentioned in the previous sections, this study uses two databases maintained by BTS: DB1B and T-100.

DB1B provides detailed information on flights operated by US air carriers and is being published quarterly since 1993. The database is generated from a sample of airline tickets and contains data on the origin, destination, and other itinerary details of passengers such as fare information and flight routes. It is used to analyze trends in air travel, monitor airline market shares, and evaluate air transport demand. The DB1B consists of three data tables: DB1BCoupon, DB1BMarket, and DB1BTicket. The DB1BCoupon is the most microscopic data table in DB1B as it encompasses information about individual flight segments within a passenger's itinerary (i.e. single flight leg) such as operating airline, origin and destination airports, number of passengers, fare class, coupon type, trip break indication, and distance. The DB1BMarket table contains aggregated data at the market-level, such as the reporting carrier, origin, and destination airport, prorated market fare, number of market coupons, market miles flown, and carrier change indicators. The DB1BTicket table contains information about individual passenger tickets (i.e. itineraries) such as ticket numbers, the number of coupons within a ticket, ticket fares, the ticketing carrier, passenger counts, and other ticket-related details. Each ticket may include one or more flight legs. These tables are interconnected in various fields and can be used together to analyze and understand air travel patterns, fare structures, passenger flows, and market dynamics in the airline industry.

T-100 is defined as a database that contains airline market and segment (i.e. flight leg) data. The database is generated via the data collected monthly from certificated U.S. air carriers by the Office of Airline Information, BTS. The T-100 Market contains market data by carrier, origin and destination, and service class for enplaned passengers, freight, and mail. The T-100 Segment data table provides detailed information on non-stop, individual flight segments by aircraft type and service class for transported passengers, freight and mail, available capacity, scheduled departures, departures performed, aircraft hours, and load factor. This T-100 database can be used to analyze air traffic patterns, carrier market shares, and passenger, freight, and mail cargo flow within the air transportation system.

3.2. Dataset preparation

In the scope of this study, we use data from the last quarter of 2021. Sample records for DB1BTicket, DB1BMarket, and DB1BCoupon data tables are given in Table 2. First, we merged the DB1BMarket and DB1BTicket tables based on the itinerary ID information (ItinID). Samples with missing information are removed. We also removed the samples where the itinerary fare (ItinFare) is less than 50$, the distance per flight leg (MktMilesFlown/MktCoupons) is less than 100 miles, and price credibility (DolarCred) is equal to 0 since these samples are considered as reporting error and unreliable (Wang et al., 2019). Then, for each market ID (MktID), we added the market segments (i.e. flight legs) that the O—D is achieved by merging relevant data from the DB1BCoupon table based on “MktID”. A new column named “ODPairID”

Table 2
DB1B data tables sample records.

DB1BTicket										
ItinID	RoundTrip	DollarCred	FarePerMile	ItinFare	Distance	MilesFlown	...	...	...	...
202,143,381,893	1	1	0.1424	172.0	1208.0	1208.0	...	...	...	...
...	...	...	...	...	...	...	...	...	...	...
DB1BMarket										
ItinID	MktID	MktCoupons	Origin	Dest	RPCarrier	OpCarrier	MktFare	MktDistance	NonStopMiles	...
202,143,381,893	20,214,338,189,301	2	BOS	CLE	UA	UA	86.00	604.00	563.00	...
202,143,381,893	20,214,338,189,303	2	CLE	BOS	UA	UA	86.00	604.00	563.00	...
...	...	...	...	...	...	...	...	...	...	...
DB1BCoupon										
ItinID	MktID	SeqNum	Origin	Dest	Distance	...	...	...	...	...
202,143,381,893	20,214,338,189,301	1	BOS	EWR	200.00	...	...	...	...	...
202,143,381,893	20,214,338,189,301	2	EWR	CLE	404.00	...	...	...	...	...
202,143,381,893	20,214,338,189,303	3	CLE	EWR	404.00	...	...	...	...	...
202,143,381,893	20,214,338,189,303	4	EWR	BOS	200.00	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...	...

is generated based on the origin and destination cities of each sample for our O—D level prediction.

Then, the T-100 market and segment 2021 data tables are filtered to contain only the last quarter of the year. Sample records for T-100 data tables are given in Table 3. Again, a new column named "ODPairID" is defined based on the origin and destination cities of each sample. From the T-100 market data table quarterly total passenger volume of each O—D pair and each carrier in each O—D pair are calculated as new columns (Pax, CarrierPax) and merged with our dataset based on ODPairID. Following this, the T-100 segment data table is used to calculate the quarterly flight frequency in each flight segment. The flight frequency data is then merged with our dataset based on the flight segment columns of each sample. The minimum frequency for each sample (i.e. minimum frequency among the frequency of flight segments that a market ID consists of) is identified and defined as a new column (Pagoni & Psaraki-Kalouptsidi, 2016). Then, we removed samples where the quarterly passenger volume is less than 50, the minimum frequency is less than 13, and the operating carrier is different from the reporting carrier.

Competition is one of the main fare drivers as fares are expected to increase when the HHI increases (Hofer, Windle, & Dresner, 2008; Stavins, 2001). We also generated a feature for the competition factor using the Herfindhal-Hirschman Index (HHI), which is calculated based on Formula 1.

$$HHI = \sum_{c=1}^{N} \left(\frac{p_c}{P}\right)^2 \tag{1}$$

where p_c is the number of passengers carried by a carrier c, P is the total number of passengers carried between the given O—D pair, and N is the total number of carriers serving in the given O—D pair.

In some regions (origin/destination points), there are multiple airports. It is mentioned that the existence of multiple airports for an O—D pair is a determinant of fare. A multi-airport system refers to metropolitan areas served by multiple commercial airports, where passengers/airlines may choose between airports within the same region. This setup increases competition, which can lead to lower fare prices as airlines vie for customers. An example of this is the "Southwest Effect," where Southwest Airlines competes in hub-to-hub markets by serving secondary airports in multi-airport regions. For instance, competition is seen between Chicago Midway-St. Louis and Chicago O'Hare-St. Louis, or Dallas Love-Houston Hobby versus Dallas Fort Worth-Houston Bush Intercontinental (Vowles, 2006; Xiao, Dobruszkes, Mo, & Wang, 2025). Thus, we defined a binary feature (MultiAirport) to indicate whether either the origin or the destination (or both) had multiple airport options available. A multi-airport system was identified when a city code in the BTS data corresponded to more than one commercial airport code. Also, the existence of a low-cost carrier in a market is a determinant of fare (Hofer et al., 2008; Vowles, 2006). Accordingly, a binary feature that indicates the presence of a low-cost carrier is also defined. The ratio of the distance actually flown to non-stop miles is also a factor that affects fares. This factor is called circuity and is a measure of the convenience of the service between an O—D pair (Hofer et al., 2008). We calculated and included circuity in our dataset. Moreover, slot-controlled airports are generally highly congested airports. It is mentioned that fares for O—D pairs, where a slot-controlled airport is involved, tend to be higher. Hence, a binary feature that indicates whether the O—D pair consists of a slot-controlled airport or not is introduced to the dataset. Finally, our target variables are calculated as the average fare of each carrier for each O—D pair. The definitions of the features in the resulting dataset are given in Table 4.

Table 3
T-100 data tables sample records.

T-100 Market					
Month	Origin	Dest	Carrier	Pax	...
11	BOS	CLE	YX	3192	...
11	BOS	CLE	UA	145	...
11	BOS	CLE	B6	3552	...
...	...	...	...	...	...
T-100 Segment					
Month	Origin	Dest	Carrier	Departures	...
11	BOS	EWR	UA	29	...
11	EWR	CLE	UA	27	...
...	...	...	...	...	...

3.3. *Supervised ML algorithms*

Supervised learning is a type of ML approach where for each data sample, there is input and desired output, and learning the mapping from the input to the output is the task (Alpaydin, 2020). Our problem is to predict a numerical output value (i.e. target value, airfare) from a set of input values (features). These types of problems are called regression in statistics, and regression is a type of supervised learning in ML.

In the scope of this study, several well-known supervised ML algorithms are tested and compared to obtain the most effective prediction for the given dataset. Seven approaches are applied: linear regression, ridge regression, k-nearest Neighbors, decision tree, random forest, gradient boosting, and an artificial neural network.

Table 4
Features in the initial dataset.

Name	Description
MktCoupons	Number of flight segments involved in achieving the O—D pair.
OriginCityMarketID	Origin city identifier.
DestCityMarketID	Destination city identifier.
OriginAirportID	Origin airport identifier
DestAirportID	Destination airport identifier
Carrier	Airline that operated the flight.
MktMilesFlown	Distance (in miles) between the origin and destination airports.
NonStopMiles	Distance (in miles, using radian measure) of a direct non-stop flight between the origin and destination airports.
RoundTrip	Indicator of whether the itinerary that involves the O—D is a round-trip or not.
MarketShare	Percent of total passengers carried in the market by a particular airline.
MarketHHI	Measure of the degree of competition within a market.
LCCComp	Indicator of the existence of low-cost carrier serving in a market.
MultiAirport	Indicator of the existence of multiple airports within the origin, destination, or both.
Circuity	Ratio of the distance actually flown to non-stop miles.
Slot	Indicator of whether one or both airports are controlled airports.
NonStop	Indicator of whether the trip between origin and destination is achieved by a single flight leg or multiple flight legs.
Pax	Quarterly total passenger volume of each market.
CarrierPax	Quarterly total passengers carried by each carrier in each market.

Linear regression (LR) is a simple and widely used statistical method for modeling the relationship between a target and a set of features. The method aims to find the best-fitting linear equation that minimizes the sum of squared differences between the actual target values and the predicted values. Ridge regression (RR) is a regularized (l_2-norm) form of linear regression that, during model training, adds a penalty term to the coefficients. The bold part in Eq. 2 is added in RR, where n is the sample size, p is the number of features, $\beta_j, j = 0, \dots, p$ represents the coefficients,$y_i, i = 1, .., n$ is the actual target values and λ is the hyperparameter that controls the relative importance of the error on the training set and the complexity due to nonzero parameters (Alpaydin, 2020).

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=0}^{p} \beta_j x_{ij} \right)^2 + \boldsymbol{\lambda} \sum_{\mathbf{j}=\mathbf{1}}^{\mathbf{n}} \boldsymbol{\beta}_{\mathbf{j}}^{\mathbf{2}} \tag{2}$$

The k-Nearest Neighbors (KNN) regressor is a simple and popular non-parametric supervised ML algorithm. Unlike traditional regression algorithms that learn a specific mathematical function to map input features to continuous output values, the KNN takes a different approach. Instead, it memorizes the training data and makes predictions based on the similarity of new data points to the existing data. During the training phase, the KNN simply stores all the labeled data points. When a new data point is given, the KNN identifies the k training data points that are closest (most similar) to the new data point based on a chosen distance metric (e.g., Euclidean distance) (Fig. 2). The distance metric is used to measure the similarity between data points in the feature space. The average of the k-nearest neighbors' output values are then calculated. This average value is the predicted output value for the new data point (Alpaydin, 2020).

A decision tree (DT) is an efficient non-parametric method that presents a hierarchical model for supervised learning. DT creates a tree-like structure by recursively splitting the data into subsets based on the input features (Fig. 3). There are decision nodes and leaf nodes, where the former represents a decision based on a specific feature, and the latter represents a prediction value. DTs are easy to visualize, understand, and interpret as the decision process is represented as a series of simple rules (Alpaydin, 2020).

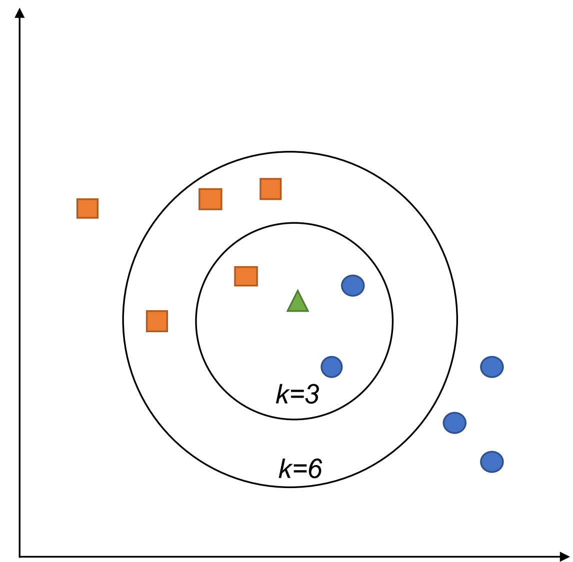

Fig. 2. KNN Illustration.

Random Forest (RF) is an ensemble method that combines multiple decision trees to enhance predictive accuracy while mitigating overfitting. The idea behind RF is to train multiple decision trees on a random subset of the data or the input features and combine their predictions to enhance the performance of the prediction (Alpaydin, 2020).

Gradient boosting (GB) is another ensemble ML technique that has gained popularity for its high predictive accuracy and efficiency. It sequentially builds multiple weak learners (decision trees) and combines their predictions to create a strong learner. GB can handle complex relationships in the data and is less prone to overfitting compared to individual decision trees (Natekin & Knoll, 2013).

Finally, the Multi-Layer Perceptron (MLP) is a type of artificial neural network and it consists of multiple layers of interconnected nodes (neurons). These layers are an input layer, one or more hidden layers, and an output layer, where each layer (except the output layer) has an associated activation function to produce an output (Fig. 4). The connections between neurons have associated weights, which are learned during the training process. During training, forward propagation is used to make predictions, and backpropagation is used to update the model's weights based on the prediction errors (Alpaydin, 2020).

3.4. Feature selection

Feature selection is the process of selecting a subset of features that can efficiently describe the input data from the dataset. It is an important process, especially when dealing with high-dimensional data. By reducing the effect of the curse of dimensionality, this process reduces computational requirements and gives insight into data. The main goal of feature selection is to improve predictor performance. Feature selection methods are classified into filter, wrapper, and embedded methods. Filter methods act as a pre-processing step, determining the relevance of individual features based on their intrinsic characteristics without involving a specific ML model. The main idea is to identify features that have a strong correlation with the target variable or demonstrate high variance, which can be good indicators of their significance or eliminate the features that have a strong correlation among them. Wrapper methods, on the other hand, use a machine learning model as part of the feature selection process. These methods aim to find the best subset of features by employing a search algorithm to evaluate different combinations of features based on model performance. The embedded method combines the feature selection process with the model training process. It incorporates feature selection directly into the

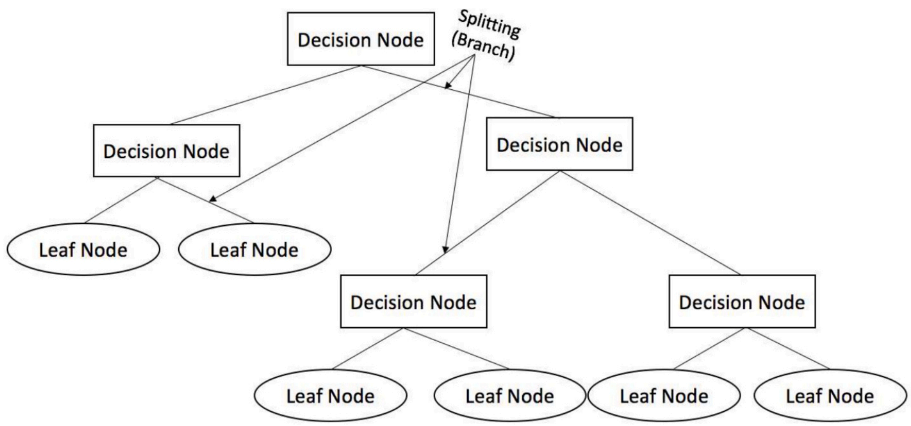

Fig. 3. DT Illustration.

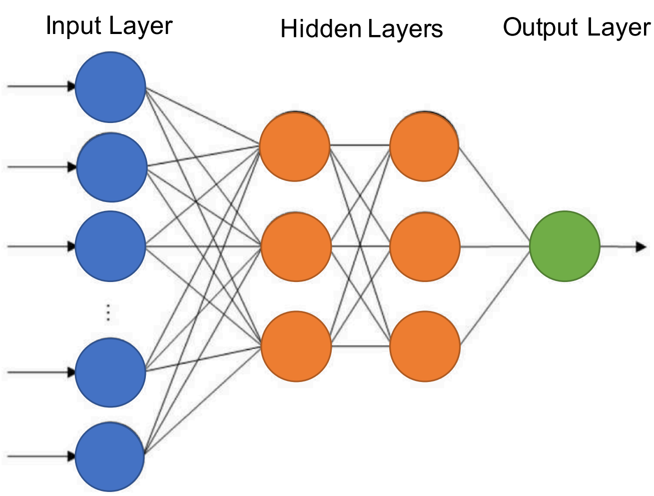

Fig. 4. MLP Illustration.

learning algorithm, optimizing both the model's performance and the feature subset simultaneously (Chandrashekar & Sahin, 2014).

One of the simplest criteria used as a filter method is the Pearson correlation coefficient. This coefficient measures the linear correlation between variables. This value ranges between −1 and 1, where 0 implies that there is no linear dependency between the variables. A value above 0.7 indicates a strong correlation (Akoglu, 2018). In this study, we used Pearson correlation coefficient criteria to identify and eliminate strongly correlated features. We used the Python seaborn library to visualize the resulting correlation matrix as a heatmap (Fig. 5). According to the results, we eliminated NonStop, NonStopMiles, MarketShare, and CarrierPax features as they are strongly correlated and hence can be represented with other features.

We then applied the Recursive Feature Elimination (RFE) technique by combining it with cross-validation. RFE is a feature selection method that iteratively removes the least significant features from the dataset according to the model's performance. Cross-validation is a resampling technique used to assess the model's performance. The dataset is divided into multiple subsets, with the model being trained and evaluated using these subsets for both training and testing in each iteration. This process allows for a more reliable estimation of the model's performance. RFE with cross-validation is a robust and effective approach as it evaluates model performance during feature selection by assessing the model's generalization performance using cross-validation at each step (Guyon, Weston, Barnhill, & Vapnik, 2002).

3.5. Experimental setup

The resulting dataset has 12 features and 1,581,278 samples. The experiments were conducted on a machine with a 2.8 GHz Intel Core i7 CPU and 4GB of RAM using Python programming language. It is observed that the time required to process the entire dataset is excessively long. Hence, we downsampled the data set to 70 % of it for the analysis. For LR, RR, DT, RF, KNN, and GB methods scikit-learn library is used, while the GB method is applied by using the XGBoost library.

To determine the performance of each regressor (ML algorithms), we applied 5-fold cross-validation using a 70/30 train-test split. We also applied GridSearchCV which is a hyper-parameter tuning technique used to systematically search for the best combination of hyper-parameters for a given model.

To evaluate the performance of airfare prediction models, two commonly used metrics to evaluate the performance of a regression model are employed: Root Mean Square Error (RMSE) and Adjusted R-squared (R^2_{adj}). RMSE measures the average squared difference between the predicted values and the actual observed values of a dataset. It penalizes larger prediction errors more than smaller errors due to the squaring operation. A lower RMSE value indicates that the model's predictions are closer to the actual target values, reflecting better performance. The RMSE is calculated as follows:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{3}$$

where N is the total number of data points in the dataset, y_i is the actual target value for the i^{th} data point, and $\hat{y}_i$ is the predicted value for the i^{th} data point as given by the regression model.

R^2_{adj}, also known as the adjusted coefficient of determination, is used as a measure of the proportion of the variance in the dependent variable that is explained by the independent variables in the regression model. As a modification of R-squared, it represents the model's goodness of fit. It is useful for comparing different regression models with varying numbers of independent variables and for understanding the trade-off between model complexity and model performance. A higher R^2_{adj} value generally indicates a better-performing model with a good balance between explanatory power and model complexity. R^2_{adj} is calculated as follows:

$$R^2_{adj} = 1 - \frac{(1 - R^2) \times (n - 1)}{n - k - 1} \tag{4}$$

where R^2 is the regular R-squared (coefficient of determination), n is the number of data points (samples) in the dataset, and k is the number of features. To calculate the R^2, the below formula is used:

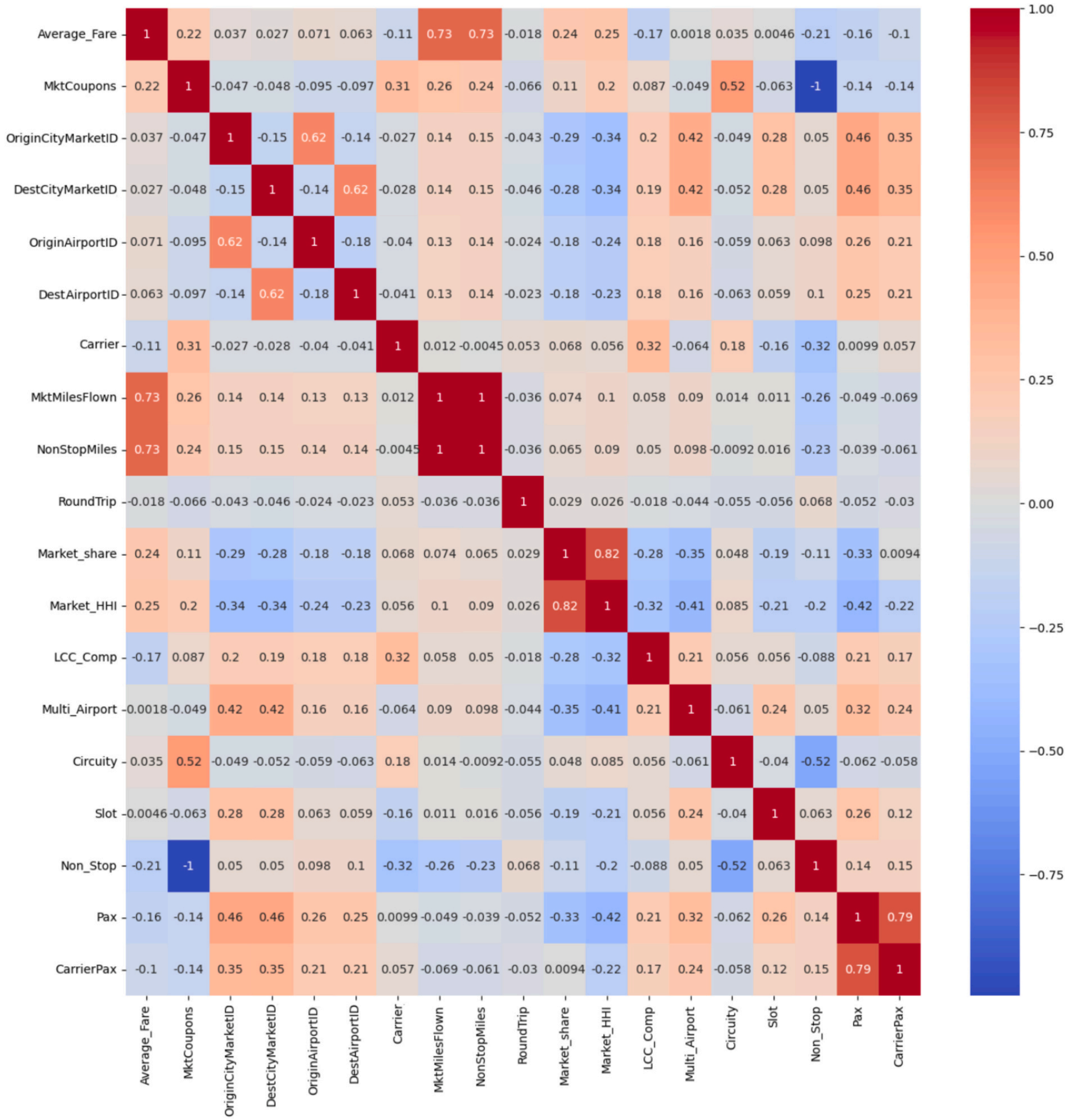

Fig. 5. Correlation heatmap.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{5}$$

where $\bar{y}$ is the mean of the actual target values.

4. Results and discussion

This section presents a performance comparison for different regression models. Moreover, a post hoc analysis of feature importance is performed and results are provided.

4.1. Model performance

As mentioned in the previous sections, seven different methods are tested for their performance in O—D level airfare prediction. We used linear regression as the baseline model. The R^2_{adj} and RMSE values, as performance indicators, for each model, are provided in Table 5.

Results show that the predictive power of most of the models is very strong. KNN, DT, and RF are the regressors with the best R^2_{adj} value, while our baseline regressor has the lowest R^2_{adj} value. Among these three, the best model was found to be the RF, as it also has the lowest RMSE value. We would also like to note that the hyper-parameters that

Table 5
Performance of different regression models.

Regressor	R^2_{adj}	RMSE
LR	0.614	45.504
RR	0.614	45.484
KNN	0.997	2.933
DT	0.997	3.073
RF	0.998	1.811
MLP	0.962	14.311

are different from the default values for the RF model are identified during the hyperparameter tuning process as max_depth: 20, n_estimators: 210. Consecutively the RF model is identified as the best-performing model and utilized in the proposed framework.

4.2. Feature importance

To gain further insight into the determinants of market-level airfares we analyze the feature importance using the RF model. The importance of each feature is evaluated by measuring the impact of feature perturbation on model performance. The results are presented in Fig. 6.

The market miles flown (MktMilesFlown) have the highest importance suggesting that the distance between the origin and destination airports is a critical factor influencing airfare. This finding aligns with common industry knowledge that longer flights typically result in higher airfares due to increased fuel and operational costs (Koopmans & Lieshout, 2016). The airline operating the flight (Carrier) also demonstrates significant importance in predicting airfare. Different carriers have distinct pricing strategies and market positions, which may impact fare levels. The total passenger volume for each market (Pax) contributes notably to airfare prediction. Higher demand often correlates with higher prices, especially during peak travel periods. Airlines may adjust prices based on passenger traffic to optimize revenue. The Herfindahl-Hirschman Index (Market_HHI), a measure of market concentration and competition, also presents a meaningful impact on airfare prediction. A higher HHI value means a higher concentration and hence less competition. In market with low competition may have higher fares due to reduced price pressure. Other features have lower importance compared to the ones mentioned in the former. The ratio of distance flown to nonstop miles (Circuity) and the number of flight segments involved in achieving the O—D pair (MktCoupons) as measures of the convenience of the service between an O—D pair are lesser but still significant determinants.

5. Conclusion

The air transportation industry plays a vital role in the global economy, with pricing and revenue management standing out as a critical aspect of airline competitiveness and profitability. Pricing and revenue management encompasses two main concerns which are the determination of airfare and seat inventory management. This study focuses on O—D level airfare prediction using machine learning techniques, aiming to enhance understanding and optimize decision-making in the context of pricing and revenue management. In this context, leveraging two publicly available databases (DB1B and T-100) provided by the United States BTS, we prepared a dataset for O—D level airfare prediction. After applying feature selection techniques, seven different models are tested and evaluated based on R^2_{adj} and RMSE performance metrics. RF is selected as the best model with high performance, R^2_{adj} and RMSE scores being 0.998 and 1.811, respectively. We also conduct an ad hoc feature importance analysis in order to investigate the impact of each feature in predicting airfare. Flight distance and the airline operating the flight are found to have predominant roles in determining airfare. These findings underline the significance of operational costs and pricing strategies in airfare setting. Market dynamics and passenger demand are also identified as significant determinants of airfare.

In that sense, this study presents an ML framework that airlines can leverage to understand the relative importance of airfare determinants, enabling them to adapt pricing strategies to maximize revenue while remaining competitive in dynamic market conditions. This study is not intended to replace traditional pricing or revenue management models but instead offers a foundation for how machine learning can support and enhance these existing systems. With airlines moving toward continuous pricing models instead of fixed-fare classes, the integration of machine learning could significantly improve decision-making. The results obtained show strong predictive results using publicly available information, suggesting that with more detailed and global data, machine learning could be even more useful. However, there are some limitations due to the nature of the databases used. For instance, the BTS databases primarily focus on flights within the United States. Airline industry dynamics and market conditions can vary and a dataset that has a more global coverage can better capture the landscape of airfare prediction. Also, the fare class information is not included in this study as DB1B specifically declared that data regarding fare class may not follow the same standard and is not recommended for analysis. Finally,

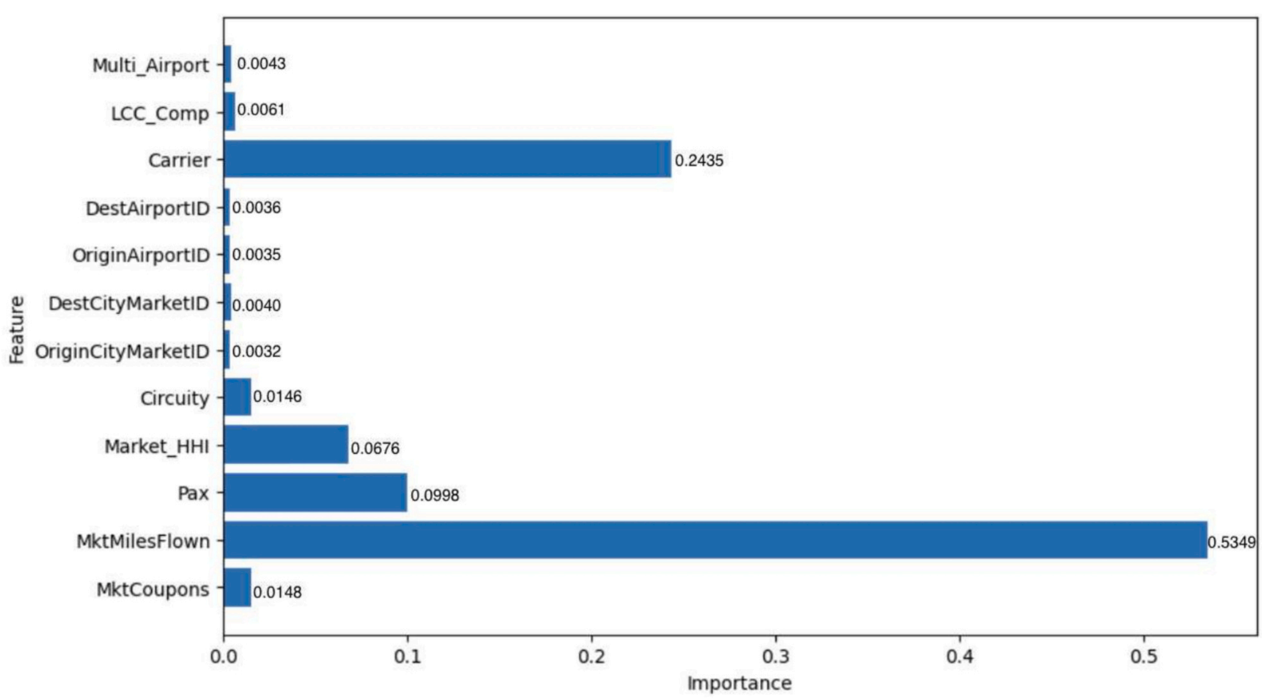

Fig. 6. Feature importances.

the data does not include any information regarding the time of the flights, which may be an important determinant of airfare, as different days of the week and different times of day may have different demands. It is worth investigating the temporal and seasonal variations with a global dataset in future research. This could provide a broader perspective on airfare dynamics influenced by varying market conditions and regulatory environments. Moreover, conducting a longitudinal study to analyze long-term trends and changes in airfare dynamics could provide further insights for informed decision-making. Monitoring evolving market conditions and consumer behaviors over time could guide adaptive revenue management strategies. Overall, this study serves as a proof of concept, demonstrating how machine learning can complement and enhance traditional pricing and revenue management practices, offering valuable insights into airfare prediction that could help airlines optimize decision-making in dynamic market conditions.

CRediT author statement

K. Gulnaz Bulbul: Conceptualization, Methodology, Software, Validation, Formal analysis,

Data Curation, Writing - Original Draft, Writing - Review & Editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Grammarly for Safari extension to check for and correct grammatical and spelling errors. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRediT authorship contribution statement

K. Gülnaz Bülbül: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

None.

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Data availability

The dataset constructed and used in this study is openly available in Mendeley Data at http://doi.org/10.17632/m5mvxdx2wp.2.

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